



```
In [9]: # -----
# Step 0: Install Libraries
# -----
!pip install gradio scikit-learn pandas matplotlib wordcloud joblib

# -----
# Step 1: Load Dataset
# -----
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("toxic-comment-classification.csv")
print("First 5 rows:")
print(df.head())
print("Columns:", df.columns)

# -----
# Step 2: Preprocess Text and Labels
# -----
df['comment_text'] = df['comment_text'].str.lower().str.replace(r'\n', ' ', regex=True)
df = df.dropna()
df['label'] = df['label'].map({'not':0, 'toxic':1})

# -----
# Step 3: Split Dataset
# -----
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(df['comment_text'], df['label'], random_state=42)

# -----
# Step 4: Vectorize Text
# -----
from sklearn.feature_extraction.text import TfidfVectorizer
vectorizer = TfidfVectorizer(stop_words='english', max_features=10000)
X_train_vec = vectorizer.fit_transform(X_train)
X_test_vec = vectorizer.transform(X_test)

# -----
# Step 5: Train Model
# -----
from sklearn.linear_model import LogisticRegression
model = LogisticRegression(max_iter=1000)
model.fit(X_train_vec, y_train)

# -----
# Step 6: Evaluate Model
# -----
from sklearn.metrics import classification_report, confusion_matrix, ConfusionMatrixDisplay

y_pred = model.predict(X_test_vec)

# Classification Report
print("\nClassification Report:")
```

```

print(classification_report(y_test, y_pred, target_names=['Not Toxic','Toxic'])

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
disp = ConfusionMatrixDisplay(cm, display_labels=['Not Toxic','Toxic'])
disp.plot(cmap=plt.cm.Blues)
plt.title("Confusion Matrix")
plt.show()

# Bar Chart of Metrics
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)
plt.figure(figsize=(6,4))
plt.bar(['Precision','Recall','F1-score'], [precision, recall, f1], color='sky'
plt.ylim(0,1)
plt.title("Model Performance Metrics")
plt.show()

# Word Cloud of Toxic Comments
from wordcloud import WordCloud
toxic_comments = df[df['label']==1]['comment_text']
text = " ".join(toxic_comments)
wordcloud = WordCloud(width=800, height=400, background_color='white').generate
plt.figure(figsize=(10,5))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.title("Common Words in Toxic Comments")
plt.show()

# -----
# Step 7: Save Model and Vectorizer
# -----
import joblib
joblib.dump(model, "toxic_model.pkl")
joblib.dump(vectorizer, "tfidf_vectorizer.pkl")

# -----
# Step 8: Deploy Gradio App
# -----
import gradio as gr

def predict_toxic(comment):
    if not comment.strip():
        return "Please enter a comment!"
    vec = vectorizer.transform([comment])
    pred = model.predict(vec)[0]
    return "✅ Not Toxic" if pred==0 else "❌ Toxic"

iface = gr.Interface(
    fn=predict_toxic,
    inputs=gr.Textbox(lines=4, placeholder="Type a comment here..."),
    outputs="text",

```

```
title="Toxic Comment Detection",  
description="Enter a comment to check if it's toxic or not.\n✅ Not Toxic  
)  
iface.launch(share=True)
```

Requirement already satisfied: gradio in /usr/local/lib/python3.12/dist-packages (5.49.1)

Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)

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Requirement already satisfied: httpx<1.0,>=0.24.1 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.28.1)

Requirement already satisfied: huggingface-hub<2.0,>=0.33.5 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.36.0)

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Requirement already satisfied: pillow<12.0,>=8.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (11.3.0)

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Requirement already satisfied: python-multipart>=0.0.18 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.0.20)

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Requirement already satisfied: fsspec in /usr/local/lib/python3.12/dist-packages (from gradio-client==1.13.3->gradio) (2025.3.0)

Requirement already satisfied: websockets<16.0,>=13.0 in /usr/local/lib/python3.12/dist-packages (from gradio-client==1.13.3->gradio) (15.0.1)

Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.16.3)

Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)

Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)

Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)

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Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.60.1)

Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.4.9)

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Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.12/dist-packages (from anyio<5.0,>=3.0->gradio) (3.11)

Requirement already satisfied: sniffio>=1.1 in /usr/local/lib/python3.12/dist-packages (from anyio<5.0,>=3.0->gradio) (1.3.1)

Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from fastapi<1.0,>=0.115.2->gradio) (0.0.4)

Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1.0,>=0.24.1->gradio) (2025.10.5)

Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1.0,>=0.24.1->gradio) (1.0.9)

Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1.0,>=0.24.1->gradio) (0.16.0)

Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=0.33.5->gradio) (3.20.0)

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Requirement already satisfied: tqdm>=4.42.1 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=0.33.5->gradio) (4.67.1)

Requirement already satisfied: hf-xet<2.0.0,>=1.1.3 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=0.33.5->gradio) (1.2.0)

Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic<2.12,>=2.0->gradio) (0.7.0)

Requirement already satisfied: pydantic-core==2.33.2 in /usr/local/lib/python3.12/dist-packages (from pydantic<2.12,>=2.0->gradio) (2.33.2)

Requirement already satisfied: typing-inspection>=0.4.0 in /usr/local/lib/python3.12/dist-packages (from pydantic<2.12,>=2.0->gradio) (0.4.2)

Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)

Requirement already satisfied: click>=8.0.0 in /usr/local/lib/python3.12/dist-packages (from typer<1.0,>=0.12->gradio) (8.3.0)

Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer<1.0,>=0.12->gradio) (1.5.4)

Requirement already satisfied: rich>=10.11.0 in /usr/local/lib/python3.12/dist-packages (from typer<1.0,>=0.12->gradio) (13.9.4)

Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=10.11.0->typer<1.0,>=0.12->gradio) (4.0.0)

Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=10.11.0->typer<1.0,>=0.12->gradio) (2.19.2)

Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests->huggingface-hub<2.0,>=0.33.5->gradio) (3.4.4)

Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests->huggingface-hub<2.0,>=0.33.5->gradio) (2.5.0)

Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=10.11.0->typer<1.0,>=0.12->gradio) (0.1.2)

First 5 rows:

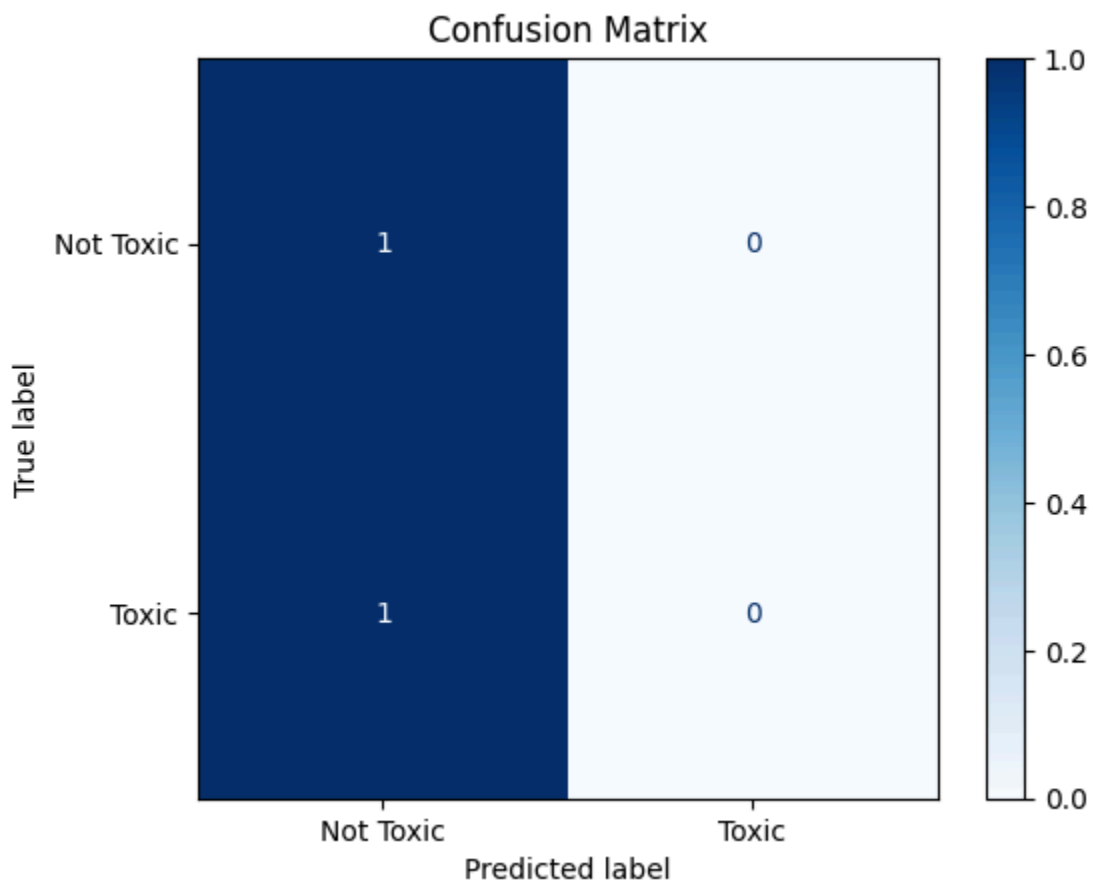
	comment_text	label
0	I love this product!	not
1	You're an idiot!	toxic
2	Great work, well done.	not
3	This is the worst thing ever.	toxic
4	Amazing experience, thank you!	not

Columns: Index(['comment_text', 'label'], dtype='object')

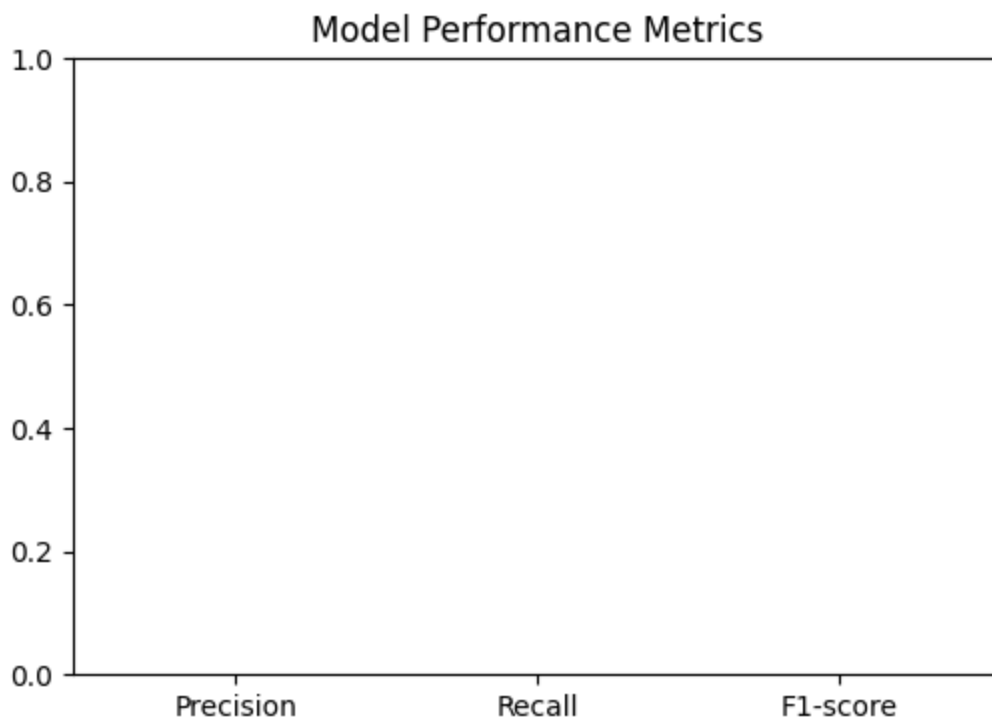
Classification Report:

	precision	recall	f1-score	support
Not Toxic	0.50	1.00	0.67	1
Toxic	0.00	0.00	0.00	1
accuracy			0.50	2
macro avg	0.25	0.50	0.33	2
weighted avg	0.25	0.50	0.33	2

```
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:156
5: UndefinedMetricWarning: Precision is ill-defined and being set to 0.0 in lab
els with no predicted samples. Use `zero_division` parameter to control this be
havior.
    _warn_prf(average, modifier, f"{metric.capitalize()} is", len(result))
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```
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:156
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no predicted samples. Use `zero_division` parameter to control this behavior.
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```



Colab notebook detected. To show errors in colab notebook, set debug=True in launch()

* Running on public URL: <https://b15d21c9264ca19435.gradio.live>

This share link expires in 1 week. For free permanent hosting and GPU upgrades, run ``gradio deploy`` from the terminal in the working directory to deploy to Hugging Face Spaces (<https://huggingface.co/spaces>)

